

Chapter 12: Reactions Involving Pure Condensed Phases and a Gaseous Phase Part II

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn
<http://teacher.nwpu.edu.cn/kehua>

Outlines

12.1 Introduction

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

12.4 Ellingham Diagrams

12.5 The Effect of Phase Transformations

12.6 The Oxides of Carbon

12.7 Graphical Representation of Equilibria in the System Metal-Carbon Oxygen

12.8 Summary

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

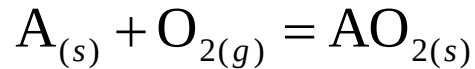
12.8 Summary

12.5 The Effect of Phase Transformations

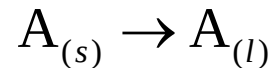
- ✓ In the previous section, it was stated that the variation of ΔG° with temperature can be approximated by a straight line only over ranges of temperature in which no change of phase of a reactant or product occurs.
- ✓ However, the enthalpy of a high-temperature phase (e.g., the liquid phase) exceeds that of a low-temperature phase (e.g., the solid phase) by the latent enthalpy of the phase change.
- ✓ Similarly, the entropy of the higher-temperature phase exceeds that of the lower-temperature phase.
- ✓ Thus, a change in the slope of the ΔG° line (an *elbow*) occurs on an Ellingham line at the temperature of a phase change in either a reactant phase or product phase of the reaction.

12.5 The Effect of Phase Transformations

- ✓ Consider the reaction

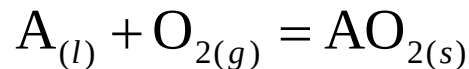


for which ΔH° is the standard enthalpy change and ΔS° is the standard entropy change. At $T_{m,A}$, the melting temperature of A, the change of phase



occurs, for which the standard enthalpy change (the enthalpy of melting) is $\Delta H_{m,A}^\circ$ and the corresponding change in entropy is $\Delta S_{m,A}^\circ$.

- ✓ Thus, for the reaction



the standard enthalpy change is $\Delta H^\circ - \Delta H_{m,A}^\circ$ and the standard entropy change is $\Delta S^\circ - \Delta S_{m,A}^\circ$.

12.5 The Effect of Phase Transformations

- ✓ Since $\Delta H_{m,A}^\circ$ and $\Delta S_{m,A}^\circ$ are positive quantities (melting is an endothermic process), then $\Delta H^\circ - \Delta H_{m,A}^\circ$ is a larger negative quantity than is ΔH° and $\Delta S^\circ - \Delta S_{m,A}^\circ$ is a larger negative quantity than is ΔS° .
- ✓ Consequently, the Ellingham line for the oxidation of liquid A to form solid AO_2 has a greater slope than the corresponding line for the oxidation of solid A, and the line contains an *elbow upward* at $T_{m,A}$.
- ✓ This is shown in Figure 12.9a. The line does not contain a discontinuity, since, at $T_{m,A}$, $G_{\text{A(s)}}^\circ = G_{\text{A(l)}}^\circ$

12.5 The Effect of Phase Transformations

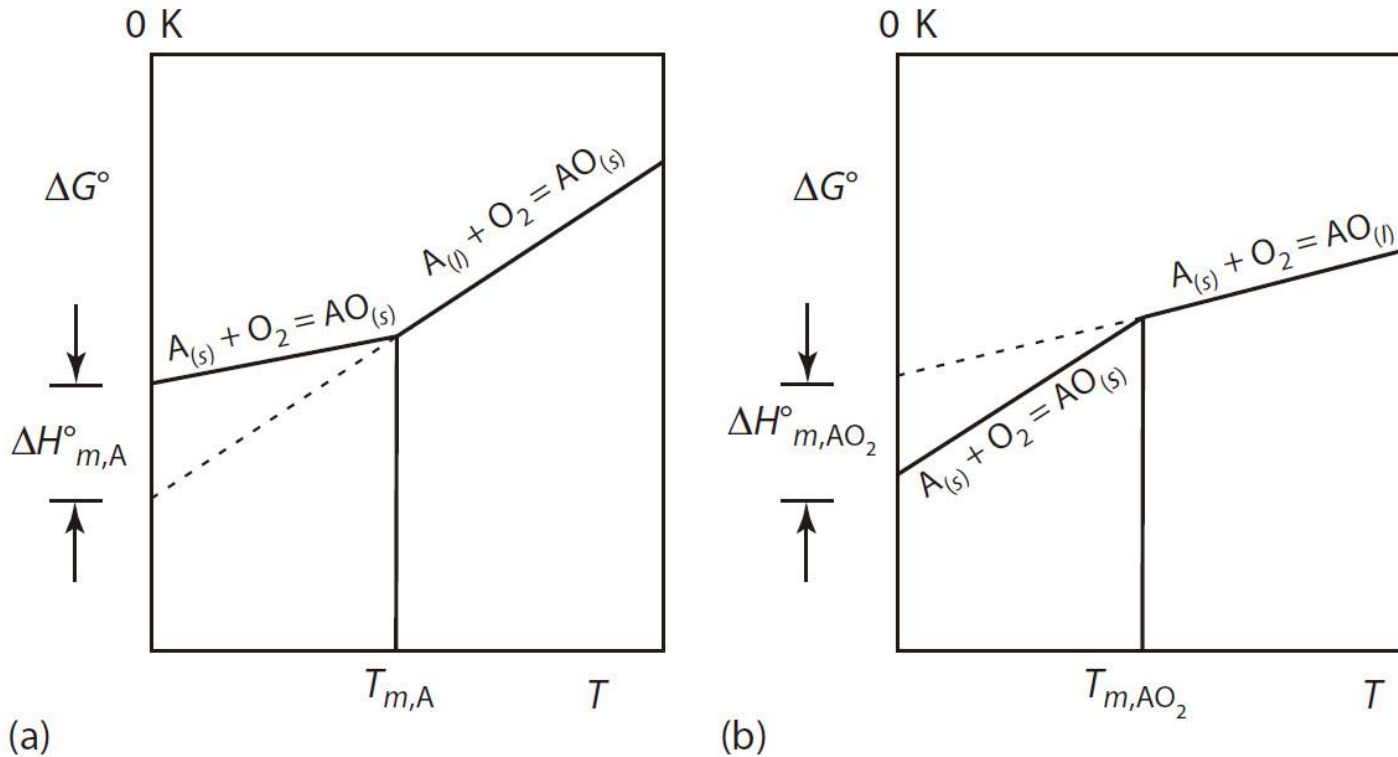
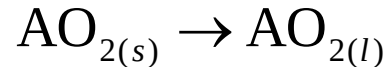


Figure 12.9 (a) The effect of melting of the metal on the Ellingham line for the oxidation of the metal. (b) The effect of melting of the metal oxide on the Ellingham line for the oxidation of the metal.

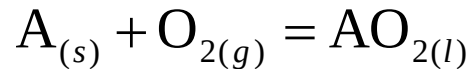
12.5 The Effect of Phase Transformations

- ✓ If the melting temperature of the oxide, T_{m, AO_2} , is lower than the melting temperature of the metal, then, at T_{m, AO_2} , the change of phase



occurs, for which the standard enthalpy and entropy changes are, respectively, $\Delta H_{m, \text{AO}_2}^\circ$ and $\Delta S_{m, \text{AO}_2}^\circ$.

- ✓ Thus, for the reaction



the standard enthalpy change is $\Delta H^\circ + \Delta H_{m, \text{AO}_2}^\circ$ and the standard entropy change is $\Delta S^\circ + \Delta S_{m, \text{AO}_2}^\circ$, both of which are less than the corresponding quantities ΔH° and ΔS° .

- ✓ In this case, the Ellingham line for oxidation of the solid metal to produce the liquid oxide has a lower slope than the line for oxidation of the solid metal to the solid oxide, and thus, as shown in Figure 12.9b, the Ellingham line has an *elbow downward* at T_{m, AO_2} .

12.5 The Effect of Phase Transformations

- ✓ In brief, if $T_{m,A} < T_{m,AO_2}$, the Ellingham line is as shown in Figure 12.10a, and, if $T_{m,A} > T_{m,AO_2}$, the line is as shown in Figure 12.10b.

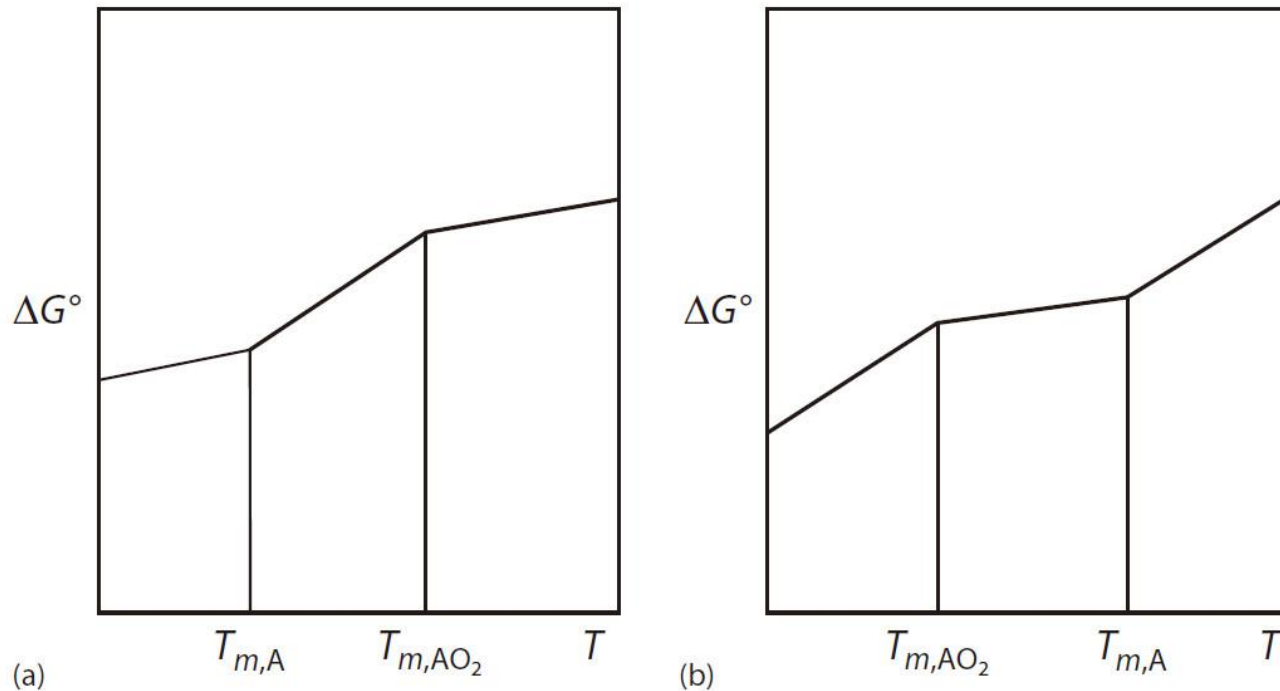


Figure 12.10 Illustration of the effects of phase changes of the reactants and products of a reaction on the Ellingham line for the reaction.

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

12.8 Summary

12.5.1 Example of the Oxidation of Copper

- ✓ Copper is a metal which melts at a lower temperature than its lowest oxide Cu_2O . The standard Gibbs free energy change for the oxidation of solid copper to form solid cuprous oxide in the range of temperature 298 K to $T_{m,\text{Cu}}$ is

$$\Delta G^\circ = -338,900 - 14.2T \ln T + 247T \text{ J} \quad (\text{i})$$

and for the oxidation of liquid copper to form solid cuprous oxide in the range of temperature $T_{m,\text{Cu}}$ to 1503 K,

$$\Delta G^\circ = -390,800 - 14.2T \ln T + 285.3T \text{ J} \quad (\text{ii})$$

12.5.1 Example of the Oxidation of Copper

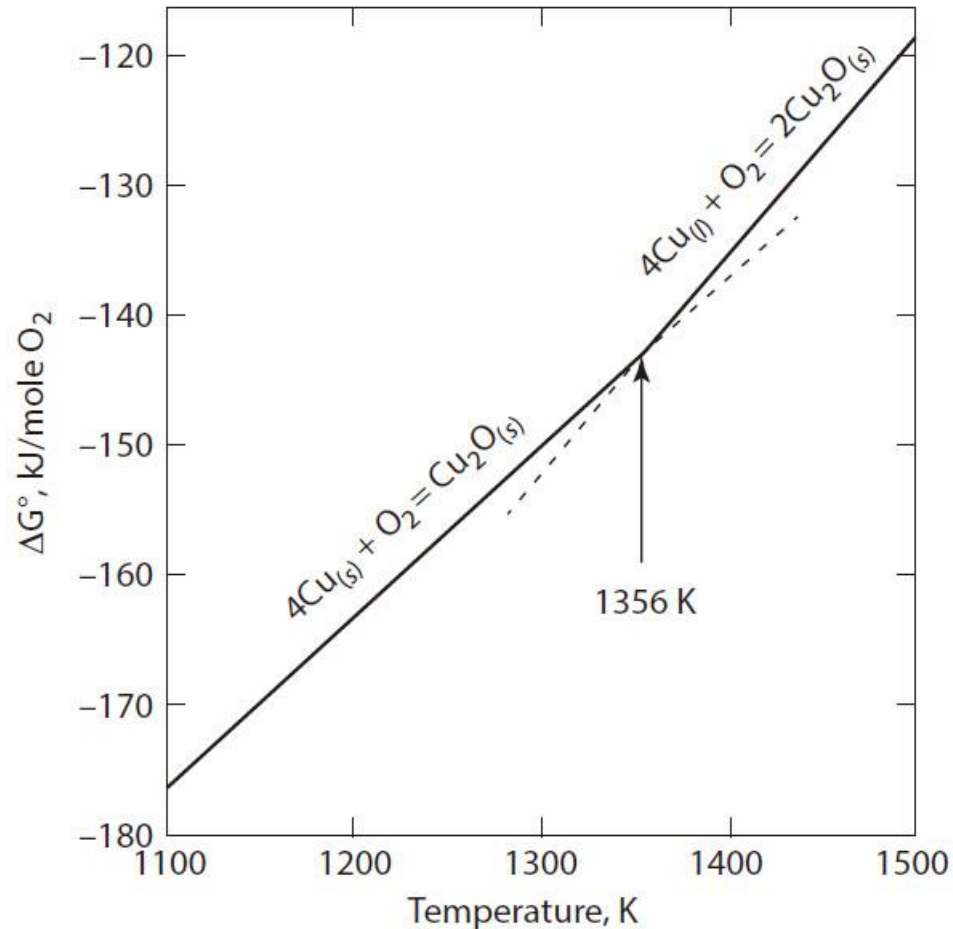


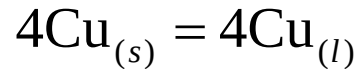
Figure 12.11 The Ellingham line for the oxidation of copper.

12.5.1 Example of the Oxidation of Copper

- ✓ These two lines, which are drawn in Figure 12.11, intersect at 1356 K, which is the melting temperature of copper. $\Delta G_{(i)}^{\circ} - \Delta G_{(ii)}^{\circ}$ gives

$$\Delta G = 51,900 - 38.3T \text{ J}$$

for the phase change



or, for the melting of 1 mole of Cu,

$$\Delta G_{m,\text{Cu}}^{\circ} = 12,970 - 9.58T \text{ J}$$

from which

$$\Delta H_{m,\text{Cu}}^{\circ} = 12,970 \text{ J}$$

and

$$\Delta S_{m,\text{Cu}}^{\circ} = 9.58 \text{ J/K}$$

- ✓ Thus, at $T_{m,\text{Cu}}$, the slope of the Ellingham line for the oxidation of Cu increases by 9.58 J/K.

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

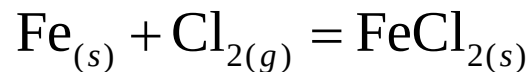
Metal-Carbon-Oxygen

12.8 Summary

12.5.2 Example of the Chlorination of Iron

- ✓ Since FeCl_2 boils at a lower temperature than the melting temperature of Fe, the Ellingham diagram for the chlorination of Fe shows elbows downward at the melting temperature of FeCl_2 and at the boiling temperature of FeCl_2 .

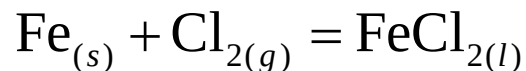
- ✓ For



$$\Delta G^\circ = -346,300 - 12.68T \ln T + 212.9T \text{ J} \quad (\text{iii})$$

in the range 298 K to T_{m,FeCl_2} .

- ✓ For

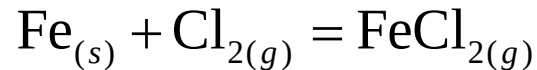


$$\Delta G^\circ = -286,400 + 63.68T \text{ J} \quad (\text{iv})$$

in the range T_{m,FeCl_2} to T_{b,FeCl_2} .

12.5.2 Example of the Chlorination of Iron

✓ For



$$\Delta G^\circ = -105,600 + 41.8T \ln T - 375.1T \text{ J} \quad (\text{v})$$

in the range T_{b,Cl_2} to $T_{m,\text{Fe}}$.

✓ Lines (iii), (iv), and (v) are shown in Figure 12.12, which shows that

$$T_{m,\text{FeCl}_2} = 969 \text{ K} \quad \text{and} \quad T_{b,\text{FeCl}_2} = 1298 \text{ K}$$

✓ For $\text{FeCl}_{2(s)} \rightarrow \text{FeCl}_{2(l)}$, $\Delta G_{(iv)}^\circ - \Delta G_{(iii)}^\circ$ gives

$$\Delta G_{m,\text{FeCl}_2}^\circ = 59,900 + 12.68T \ln T - 149.0T \text{ J}$$

12.5.2 Example of the Chlorination of Iron

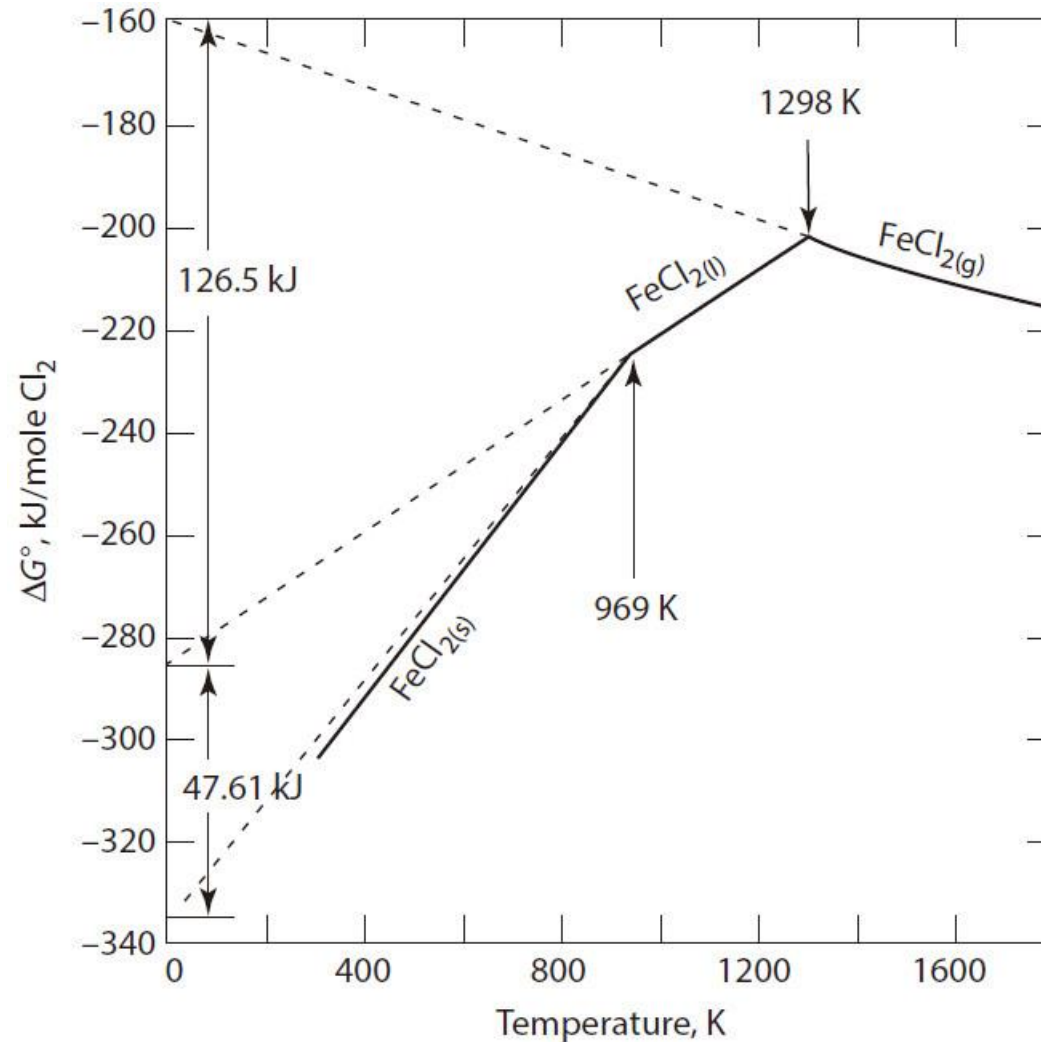


Figure 12.12 The Ellingham diagram for the chlorination of iron.

12.5.2 Example of the Chlorination of Iron

✓ Thus,

$$\Delta H_{m,\text{FeCl}_2}^{\circ} = -T^2 \left[\frac{\partial (\Delta G_{m,\text{FeCl}_2}^{\circ} / T)}{\partial T} \right] = 59,900 - 12.68T \text{ J}$$

which, at 969 K, gives $\Delta H_{m,\text{FeCl}_2}^{\circ} = 47,610 \text{ J}$.

$$\Delta S_{m,\text{FeCl}_2}^{\circ} = \frac{\partial \Delta G_{m,\text{FeCl}_2}^{\circ}}{\partial T} = -12.68 \ln T - 12.68 + 149.0 \text{ J/K} = 49.13 \text{ J/K at 969 K}$$

or, alternatively,

$$\Delta S_{m,\text{FeCl}_2}^{\circ} = \frac{\Delta H_{m,\text{FeCl}_2}^{\circ}}{T_{m,\text{FeCl}_2}} = \frac{47,610}{969} = 49.13 \text{ J/K}$$

✓ Thus, the difference in slope between lines (iii) and (iv) at 969 K is 49.13 J/K, and the difference between the tangential intercepts of the slopes of the two lines at 969 K, with the $T = 0$ axis, is 47,610 J.

12.5.2 Example of the Chlorination of Iron

✓ Similarly, $\Delta G_{(v)}^{\circ} - \Delta G_{(iv)}^{\circ}$ gives

$$\Delta G_{b, \text{FeCl}_2}^{\circ} = 180,800 + 41.8T \ln T - 438.8T \text{ J}$$

✓ Thus,

$$\Delta H_{b, \text{FeCl}_2}^{\circ} = -T^2 \left[\frac{\partial (\Delta G_{b, \text{FeCl}_2}^{\circ} / T)}{\partial T} \right] = 180,800 - 41.8T \text{ J} = 126,500 \text{ J at } 1298 \text{ K}$$

and

$$\Delta S_{b, \text{FeCl}_2}^{\circ} = \frac{126,500}{1298} = 97.46 \text{ J/K at } 1298 \text{ K}$$

✓ Thus, the change in slope between lines (iv) and (v) at 1298 K is 97.46 J/K, and the difference between the tangential intercepts is 126,500 J.

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

12.8 Summary

12.6 The Oxides of Carbon

- ✓ Carbon forms two gaseous oxides, CO and CO₂ , according to

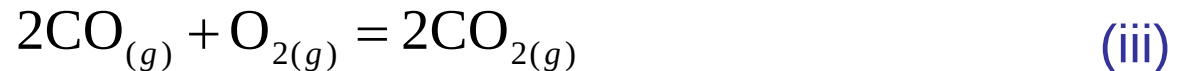


for which $\Delta G_{(i)}^{\circ} = -394,100 - 0.84T \text{ J}$, and



for which $\Delta G_{(ii)}^{\circ} = -223,400 - 175.3T \text{ J}$.

- ✓ Combination of reactions (i) and (ii) gives



for which $\Delta G_{(iii)}^{\circ} = 2\Delta G_{(i)}^{\circ} - \Delta G_{(ii)}^{\circ} = -564,800 + 173.62T \text{ J}$.

12.6 The Oxides of Carbon

- ✓ The Ellingham lines for reactions (i), (ii), and (iii) are included in Figure 12.13, in which it is seen that
 - The line for reaction (iii) has a positive slope (2 moles of gas produced from 3 moles of gas, $\Delta S_{(iii)}^{\circ} = -173.62 \text{ J/K}$).
 - The line for reaction (i) has virtually zero slope (1 mole of gas produced from 1 mole of gas, $\Delta S_{(i)}^{\circ} = 0.84 \text{ J/K}$).
 - The line for reaction (ii) has a negative slope (2 moles of gas produced from 1 mole of gas, $\Delta S_{(ii)}^{\circ} = 175.3 \text{ J/K}$).
- ✓ Consider the equilibrium



for which $\Delta G_{(iv)}^{\circ} = \Delta G_{(ii)}^{\circ} - \Delta G_{(i)}^{\circ} = 170,700 - 174.5T \text{ J}$. Reaction (iv) is known as the *Boudouard reaction* (Octave Leopold Boudouard, 1872–1923) and is important in the reduction of the oxides of iron in the blast furnace.

12.6 The Oxides of Carbon

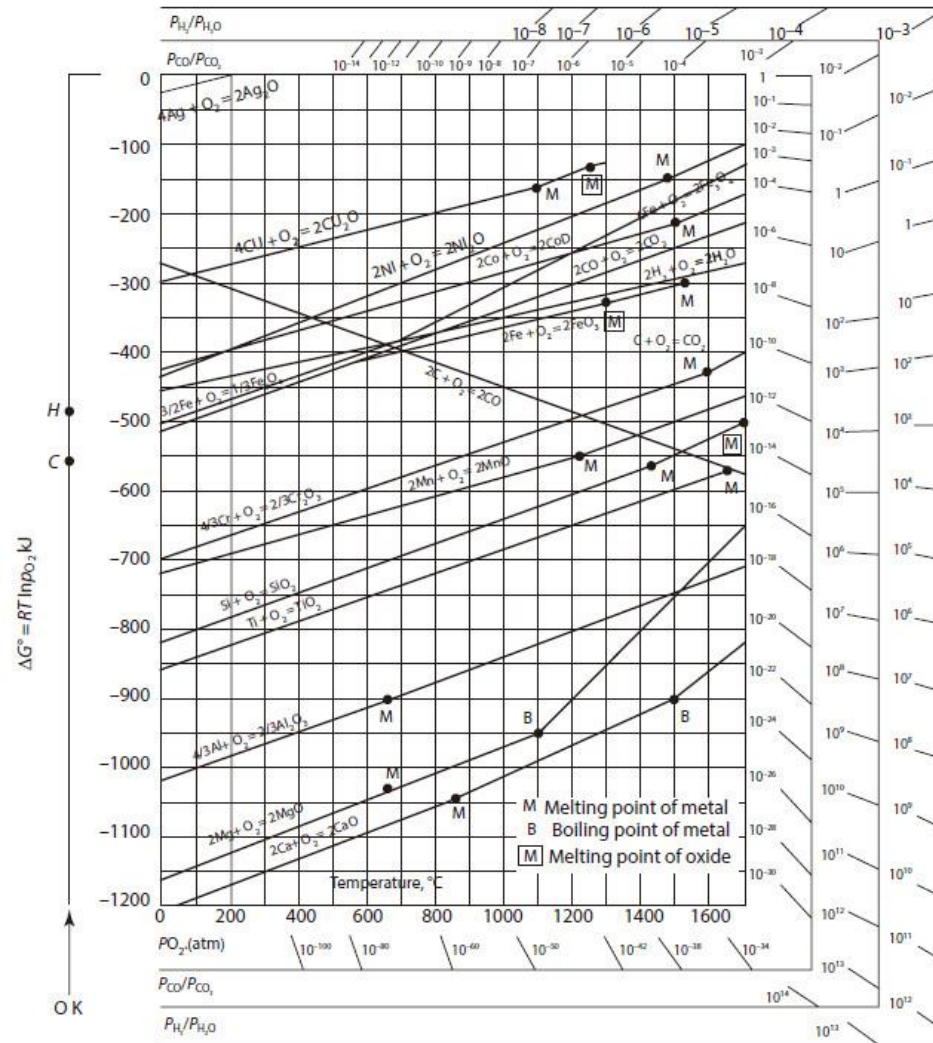


Figure 12.13 The Ellingham diagram for selected oxides.

12.6 The Oxides of Carbon

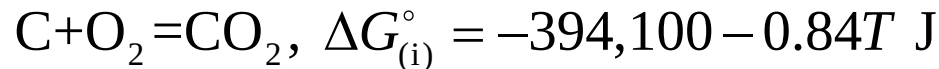
- ✓ The value of $\Delta G_{(iv)}^{\circ} = 0$ at $T = 978 \text{ K}$ ($705 \text{ }^{\circ}\text{C}$), the temperature at which the Ellingham lines for reactions (i) and (ii) intersect one another.
- ✓ At this temperature, CO and CO₂, in their standard states (i.e., both at 1 atm pressure), are in equilibrium with solid graphite, and the total pressure of the system is 2 atm.
- ✓ Since reaction equilibria are normally considered for systems under a total pressure of 1 atm, it is instructive to calculate the temperature at which CO and CO₂, each at a pressure of 0.5 atm, are in equilibrium with solid C.
- ✓ Consideration of Le Chatelier's principle indicates whether this temperature is higher or lower than 978 K.

12.6 The Oxides of Carbon

- ✓ For reaction (iv), $\Delta G_{978\text{ K}}^{\circ} = 0 = -RT \ln K_p = -RT \ln \left(p_{\text{CO}}^2 / p_{\text{CO}_2} \right)$; that is, at 978 K, $K_p = 1$, and hence $p_{\text{CO}_2} = p_{\text{CO}} = 1$ and $p_{\text{total}} = 2$ atm.
- ✓ If the pressure of the system is decreased to 1 atm, then, as K_p is independent of pressure (remaining equal to unity) p_{CO} becomes greater than p_{CO_2} ; that is, the equilibrium shifts toward the CO side, as is predicted by Le Chatelier's principle.
- ✓ Since $\Delta H_{(\text{iv})}^{\circ} = +170,700$ J, reaction (iv) is endothermic, and thus, from Le Chatelier's principle, as a decrease in temperature shifts the equilibrium in that direction which involves an evolution of heat, a decrease in temperature shifts the equilibrium toward the C + CO₂ side.
- ✓ Thus, if it is required to decrease the pressure of the system from 2 to 1 atm and, at the same time, maintain $p_{\text{CO}} = p_{\text{CO}_2}$, the temperature of the system must be decreased.

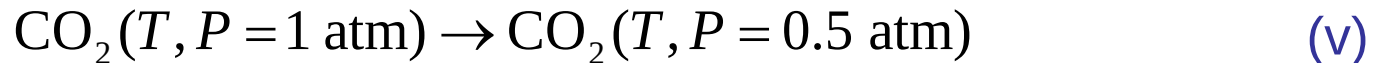
12.6 The Oxides of Carbon

- ✓ The temperature required for $p_{\text{CO}} = p_{\text{CO}_2}$ is calculated as follows.
- ✓ For reaction (i):

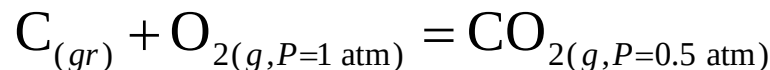


这个公式和课本不同

- ✓ If the pressure of the CO_2 , which is produced at 1 atm, is decreased to 0.5 atm, then, for the change of state



the decrease in Gibbs free energy is $\Delta G_{(v)} = RT \ln 0.5$, and hence, for the reaction

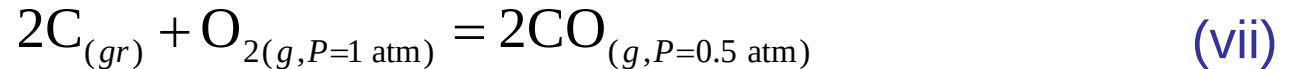


$$\Delta G_{(iv)} = \Delta G_{(i)}^\circ + \Delta G_{(v)} = -394,100 - 0.84T + RT \ln 0.5 \text{ J} \quad (\text{vi})$$

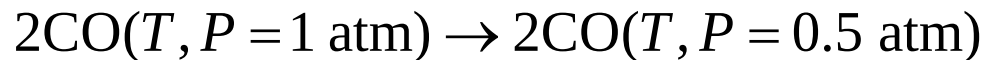
12.6 The Oxides of Carbon

- ✓ This line is obtained on the Ellingham diagram by rotating the line for reaction (i) clockwise about its point of intersection with the $T = 0$ axis until, at the temperature T , the vertical separation between line (i) and line (vi) is $RT \ln 0.5$. This is illustrated in Figure 12.14.

- ✓ Similarly, the Ellingham line for the reaction



is obtained as the sum of $\Delta G_{(ii)}^\circ$ and ΔG for the change of state



- ✓ That is,

$$\Delta G_{(vii)} = -223,400 - 175.3T + 2RT \ln 0.5 \text{ J}$$

- ✓ This line is obtained by rotating the line for reaction (ii) clockwise about its point of intersection with the $T = 0$ axis until, at any temperature T , the vertical separation between line (ii) and line (i) is $2RT \ln 0.5$.

12.6 The Oxides of Carbon

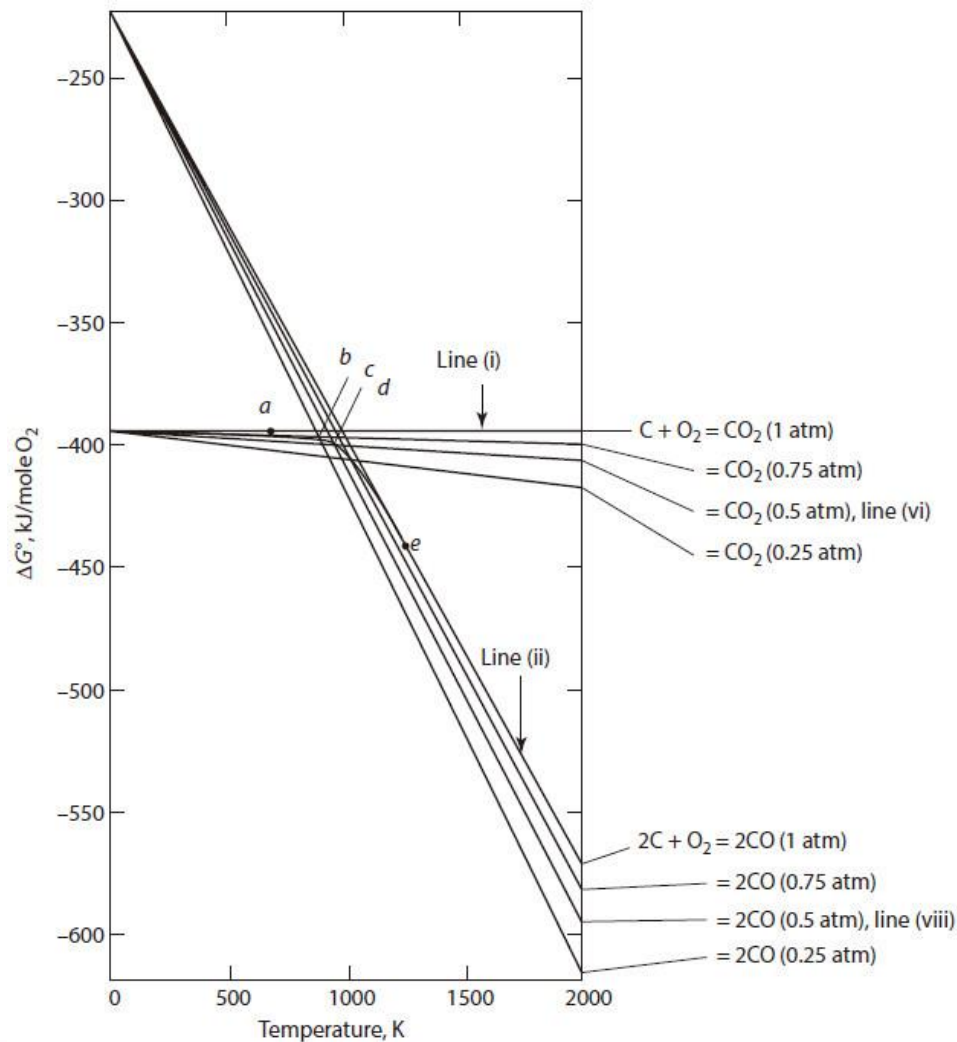
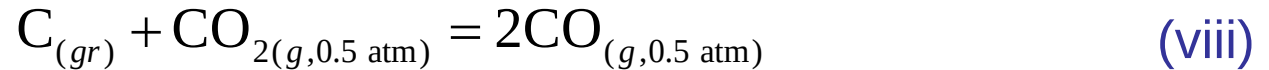


Figure 12.14 The effect of varying the pressures of the product gases of the reactions $C_{(gr)} + O_{2(g,p=1\text{ atm})} = CO_{2(g)}$ and $2C_{(gr)} + O_{2(g,p=1\text{ atm})} = 2CO_{(g)}$ on the variations of ΔG with T for the two reactions.

12.6 The Oxides of Carbon

- ✓ Combination of reactions (vi) and (vii) gives



for which

$$\Delta G_{(\text{viii})} = \Delta G_{(\text{iv})}^{\circ} + RT \ln 0.5 \text{ J}$$

- ✓ Thus, CO_2 and CO , each at 0.5 atm pressure, are in equilibrium with solid C at that temperature at which $\Delta G_{(\text{viii})} = 0$, the temperature of intersection of the lines (vi) and (vii) in Figure 12.14 (point c).
- ✓ The temperature at which CO (at 0.25 atm) and CO_2 (at 0.75 atm) are in equilibrium with solid C is obtained in a similar manner as the intersection of line (i) rotated clockwise until, at T , it has been displaced a vertical distance $RT \ln 0.75$ and line (ii) rotated clockwise until, at T , it has been displaced a vertical distance $2RT (\ln 0.25)$. This is the point b in Figure 12.14.

12.6 The Oxides of Carbon

- ✓ Similarly, the point *d* in Figure 12.14 is the temperature at which CO at 0.75 atm pressure and CO₂ at 0.25 atm pressure are in equilibrium with solid C.
- ✓ For a mixture of CO and CO₂ at 1 atm pressure in equilibrium with solid C, the variation of percent CO by volume in the gas with temperature is shown in Figure 12.15.
- ✓ Figure 12.15 includes the points *a*, *b*, *c*, *d*, and *e* drawn in Figure 12.14.

12.6 The Oxides of Carbon

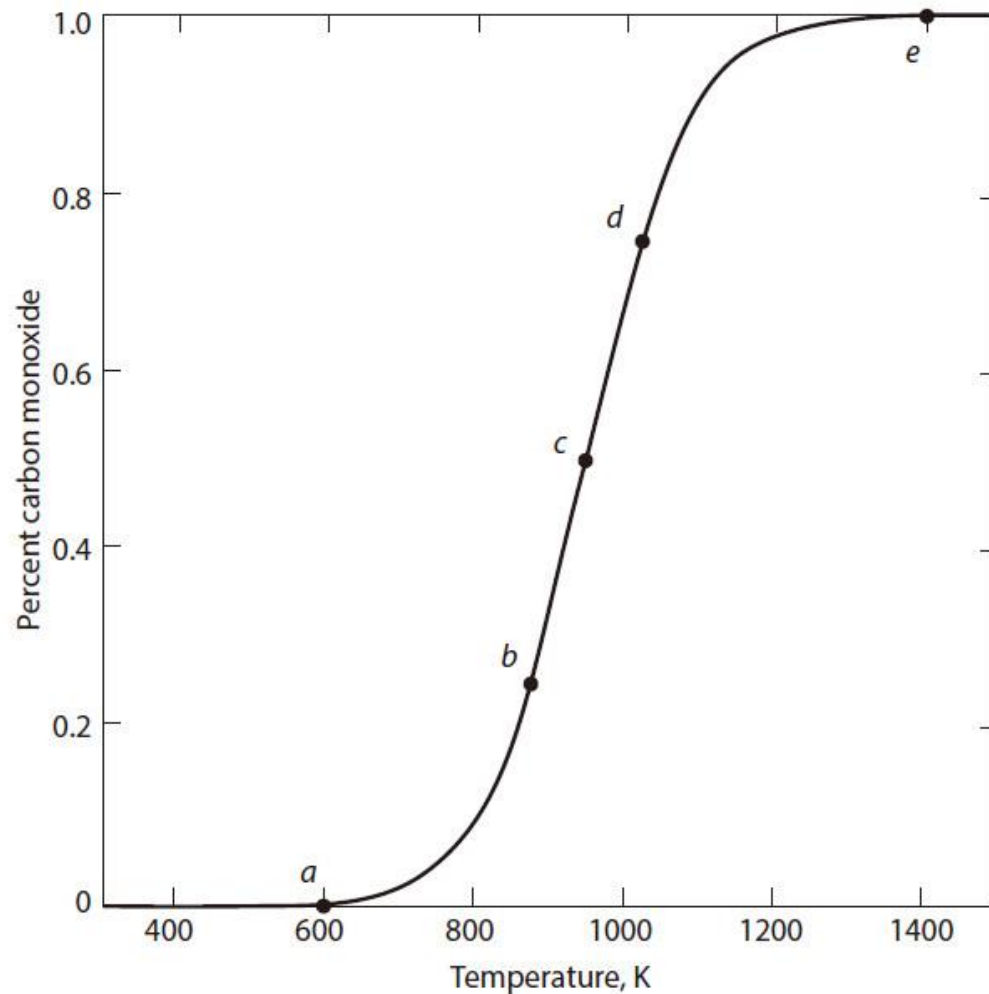


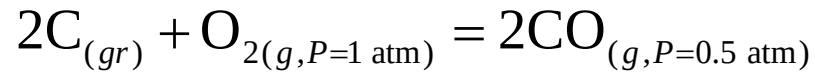
Figure 12.15 The variation, with temperature, of the composition of the CO–CO₂ gas mixture in equilibrium with solid graphite at $P_{\text{total}} = 1$ atm.

12.6 The Oxides of Carbon

- ✓ Figure 12.15 shows that, at temperatures less than 600 K, the equilibrium gas is virtually CO_2 at 1 atm pressure and, at temperatures greater than 1400 K, the equilibrium gas is virtually CO at 1 atm pressure.
- ✓ These points are, respectively, the points *a* and *e* in Figure 12.14.
- ✓ Thus, in Figure 12.14, the variation, with temperature, of the Gibbs free energy change for oxidation of solid C to produce a CO– CO_2 mixture at 1 atm pressure which is in equilibrium with solid C is given by line (i) up to the point *a*, then by the line *abcde* , and then by the line (ii) beyond the point *e* .

12.6 The Oxides of Carbon

- ✓ At any temperature T , the CO–CO₂ mixture in equilibrium with C exerts an equilibrium oxygen pressure via the equilibrium



for which

$$\Delta G_{(iii)}^\circ = -564,800 + 173.62T \text{ J} = -RT \ln \left(\frac{p_{\text{CO}_2}^2}{p_{\text{CO}}^2 p_{\text{O}_2}} \right)$$

$$= 2RT \ln \left(\frac{p_{\text{CO}}}{p_{\text{CO}_2}} \right)_{\text{eq. with C}} + RT \ln p_{\text{O}_2(\text{eq.})}$$

- ✓ Thus,

$$\ln p_{\text{O}_2(\text{eq. } T)} = \frac{-564,800}{8.3144T} + \frac{173.62}{8.3144} + 2 \ln \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right)_{\text{eq. with C}} \quad (12.12)$$

- ✓ If it is required that solid carbon be used as a reducing agent to reduce a metal oxide MO₂ at the temperature T , then $p_{\text{O}_2(\text{eq. } T)}$ in Equation 12.12 must be lower than $p_{\text{O}_2(\text{eq. } T)}$ for the equilibrium $\text{M} + \text{O}_2 = \text{MO}_2$ (see Section 12.7).

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

12.8 Summary

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

- ✓ The Ellingham line for the reaction $2\text{CO} + \text{O}_2 = 2\text{CO}_2$ is shown in Figure 12.16 as the line cs.
- ✓ Since it is the variation, with temperature, of the standard Gibbs free energy of formation, ΔG° , this line is for the reaction which produces CO_2 at 1 atm pressure from CO at 1 atm pressure and O_2 at 1 atm pressure.
- ✓ The effect of producing the CO_2 at any pressure, P , other than 1 atm (from CO and O_2 each at 1 atm) is the rotation of the Ellingham line cs about the point c, clockwise if $P < 1$ atm and anticlockwise if $P > 1$ atm.
- ✓ For the given value of P the rotation is such that, as before, at the temperature T , the vertical displacement of cs is $2RT\ln P$.
- ✓ A series of lines, radiating from the point c, can thus be drawn for different pressures of CO_2 produced from CO and O_2 , each at 1 atm.

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

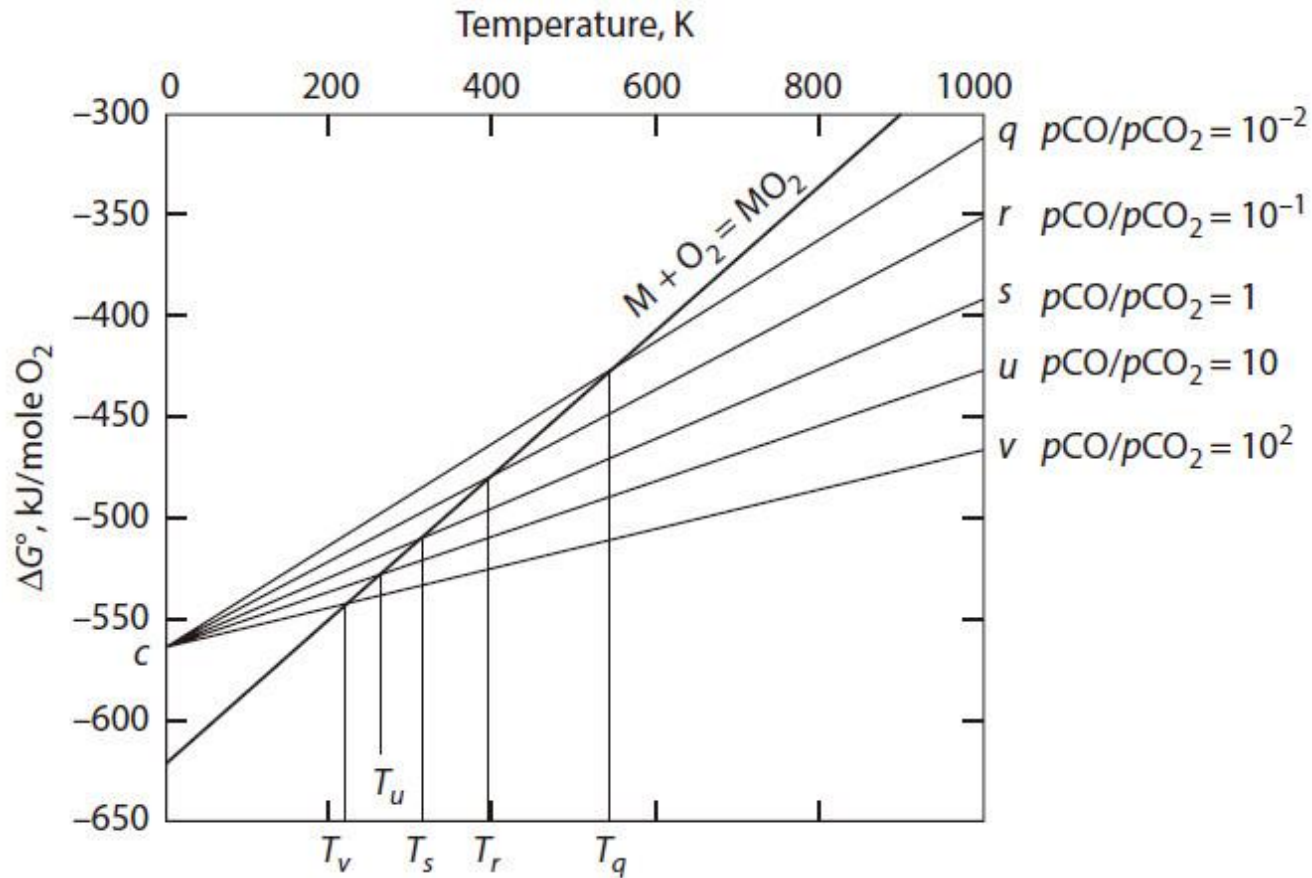


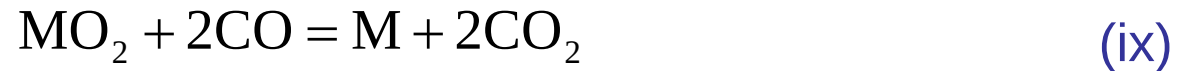
Figure 12.16 Illustration of the effect of the ratio $p_{\text{CO}_2}/p_{\text{CO}}$ in a CO_2 – CO gas mixture on the temperature at which the equilibrium $\text{M} + \text{CO}_2 = \text{MO} + \text{CO}$ is established.

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

- ✓ Figure 12.16 shows four of these lines:
 - Line cq for CO_2 produced at 10^2 atm
 - Line cr for CO_2 produced at 10 atm
 - Line cu for CO_2 produced at 0.1 atm
 - Line cv for CO_2 produced at 10^{-2} atm
- ✓ The significance of this series of lines, with respect to the possibility of using $\text{CO}-\text{CO}_2$ gas mixtures as reducing agents for the metal oxide MO_2 , is illustrated as follows. The Ellingham line for the reaction $\text{M} + \text{O}_2 = \text{MO}_2$ is drawn in Figure 12.16.

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

- ✓ This intersects the line cs at the temperature T_s , which is thus the temperature at which the standard Gibbs free energy change for reaction



is zero; that is,

$$\Delta G_{(\text{ix})}^\circ = 0 = -RT \ln \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right)^2 \quad \text{at } T_s$$

and thus,

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 1$$

- ✓ At temperatures higher than T_s , a $\text{CO}-\text{CO}_2$ mixture of $p_{\text{CO}}/p_{\text{CO}_2} = 1$ is reducing with respect to MO_2 , and at temperatures lower than T_s , it is oxidizing with respect to the metal M.
- ✓ Compare this with the previous discussion of Figure 12.2.

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

- ✓ If it is required that a $\text{CO}-\text{CO}_2$ mixture be made to be reducing with respect to MO_2 at temperatures lower than T_s , then the ratio $p_{\text{CO}}/p_{\text{CO}_2}$ must be increased to a value greater than unity.
- ✓ The Ellingham line for the reaction $\text{M} + \text{O}_2 = \text{MO}_2$ intersects the line cu at the temperature T_u , and T_u is thus the temperature at which the reaction



is at equilibrium; that is,

$$\Delta G_{(\text{x})}^\circ = \Delta G_{(\text{ix})}^\circ + 2RT \ln 0.1 \text{ at } T_u$$

- ✓ But, by definition,

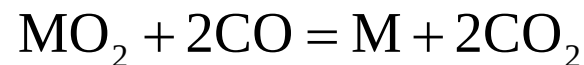
$$\Delta G_{(\text{ix})}^\circ = -RT \ln \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right)_{\text{eq}}^2$$

and thus,

$$\Delta G_{(\text{x})}^\circ = 0 = -2RT \ln \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right)_{\text{eq}} + 2RT \ln 0.1$$

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

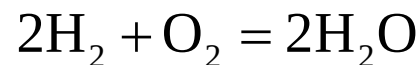
- ✓ Thus, $(p_{\text{CO}}/p_{\text{CO}_2})_{\text{eq}} = 10$ at T_u , and hence, by decreasing the temperature from T_s to T_u , the CO/CO_2 ratio must be increased from 1 to 10 in order to maintain reaction equilibrium.
- ✓ Similarly, at T_v the equilibrium CO/CO_2 ratio is 100, at T_r the equilibrium CO/CO_2 ratio is 0.1, and at T_q the equilibrium ratio is 0.01.
- ✓ Thus, a CO/CO_2 nomographic scale can be added to the Ellingham diagram, and for any reaction



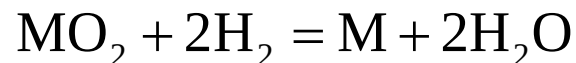
the equilibrium CO/CO_2 ratio at any temperature T is read off the nomographic scale as that point which is collinear with the point C and the point ΔG_T° , $T = T$, for the reaction $\text{M} + \text{O}_2 = \text{MO}_2$. This scale is included in Figure 12.13.

12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

- ✓ Figure 12.15 is generated by reading off the equilibrium CO/CO_2 ratios for the reaction $\text{C} + \text{O}_2 = \text{CO}_2$ up to the point *a* in Figure 12.14, then the CO/CO_2 ratios along the line *abcde*, and finally the CO/CO_2 ratios for the reaction $2\text{C} + \text{O}_2 = 2\text{CO}$ beyond the point *e*.
- ✓ In exactly the same manner, the $\text{H}_2/\text{H}_2\text{O}$ nomographic scale is added to Figure 12.13 by considering the effect of the variation of the pressure of H_2O on the reaction equilibrium



- ✓ The equilibrium $\text{H}_2/\text{H}_2\text{O}$ ratio at the temperature T for the reaction



is read off the $\text{H}_2/\text{H}_2\text{O}$ scale as the point which is collinear with the points *H* and ΔG_T° , $T = T$ for the reaction $\text{M} + \text{O}_2 = \text{MO}_2$.

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

12.8 Summary

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ The main criteria for graphical representation of equilibria in a system are
 - The amount of information provided
 - Clarity
- and both of these considerations are dependent on the coordinates chosen for use in the graphical representation. Since the nomographic scale for the ratio CO/CO_2 in Figure 12.13 shows that the range of interest of values of $p_{\text{CO}}/p_{\text{CO}_2}$ is 10^{-14} to 10^{14} , it is convenient to present this ratio on a logarithmic scale.
- ✓ Figure 12.17, which uses the coordinates $\log(p_{\text{CO}_2}/p_{\text{CO}})$, and T represents a convenient method of clearly presenting the reaction equilibrium in the carbon–oxygen and carbon–oxygen–metal systems.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

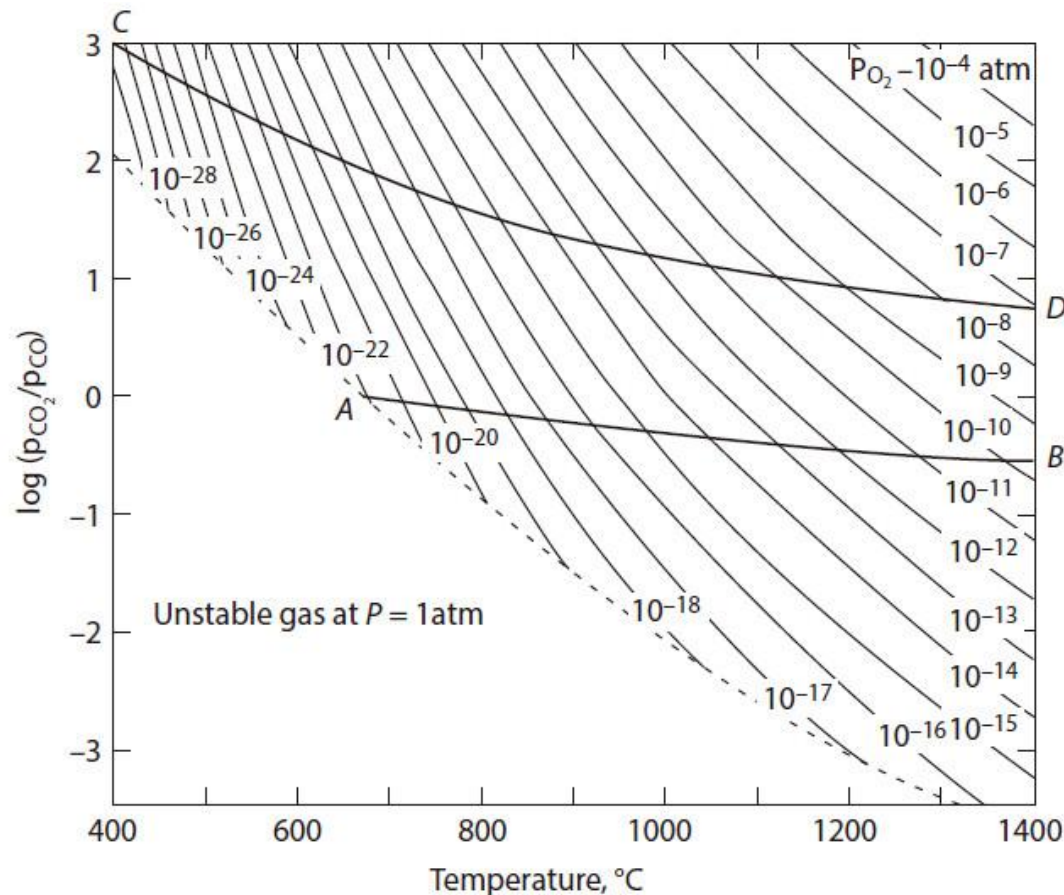
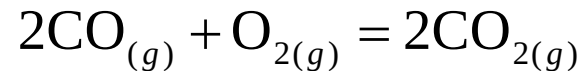


Figure 12.17 The relationship between the partial pressure of oxygen in a CO_2 -CO gas mixture and temperature. The broken line is the variation, with temperature, of the composition of the gas which is in equilibrium with graphite at 1 atm pressure. The lines AB and CD represent, respectively, the equilibria $\text{Fe} + \text{CO}_2 = \text{FeO} + \text{CO}$ and $\text{Co} + \text{CO}_2 = \text{CoO} + \text{CO}$.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ From Equation (iii), for



$$\Delta G_{(iii)}^{\circ} = -564,800 + 173.62T \text{ J} = -RT \ln \left(\frac{p_{\text{CO}_2}^2}{p_{\text{CO}}^2 p_{\text{O}_2}} \right)$$

and thus,

$$\log \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right) = \frac{1}{2} \log p_{\text{O}_2} + \frac{564,800}{2 \times 2.303 \times 8.3144T} - \frac{173.62}{2 \times 2.303 \times 8.3144} \quad (\text{xi})$$

- ✓ For any given value of p_{O_2} , this gives the variation of the given oxygen isobar with $\log(p_{\text{CO}_2}/p_{\text{CO}})$ and temperature.
- ✓ The oxygen isobars in the range 10^{-29} to 10^{-4} atm are drawn as a function of $\log(p_{\text{CO}_2}/p_{\text{CO}})$ and T in Figure 12.17.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

✓ The equilibrium



sets a lower limit on the CO_2/CO ratio which can be obtained at any temperature:

$$\Delta G_{(\text{iv})}^{\circ} = 170,700 - 174.5T \text{ J} = -RT \ln \left(\frac{p_{\text{CO}}^2}{p_{\text{CO}_2}} \right)$$

and thus, at a total pressure of 1 atm (i.e., when $p_{\text{CO}_2} = 1 - p_{\text{CO}}$),

$$\frac{p_{\text{CO}}^2}{1 - p_{\text{CO}}} = \exp \left(\frac{-170,700}{8.3144T} \right) \exp \left(\frac{174.5}{8.3144} \right) = x$$

or

$$p_{\text{CO}}^2 + p_{\text{CO}}x - x = 0$$

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

solution of which gives

$$p_{\text{CO}} = \frac{-x + \sqrt{x^2 + 4x}}{2}$$

$$p_{\text{CO}_2} = \frac{2 + x - \sqrt{x^2 + 4x}}{2}$$

and

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = \frac{2 + x - \sqrt{x^2 + 4x}}{\sqrt{x^2 + 4x} - x} \quad (\text{xii})$$

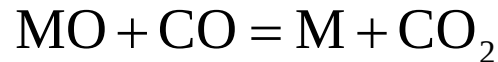
- ✓ The value of $p_{\text{CO}_2}/p_{\text{CO}}$ given by Equation (xii) is the minimum obtainable value at the temperature T .

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ If an attempt is made to mix CO and CO₂ (at $P_{\text{total}} = 1 \text{ atm}$) in a ratio $p_{\text{CO}_2} / p_{\text{CO}}$ lower than that given by Equation (xii), carbon will precipitate until, thereby, the ratio is increased to the unique value required for equilibrium with carbon at the temperature T .
- ✓ The variation of $\log(p_{\text{CO}_2} / p_{\text{CO}})_{\text{eq.C/CO/CO}_2}$ with temperature is shown as the broken line in Figure 12.17.
- ✓ The variation of the corresponding minimum values of p_{O_2} with temperature in Figure 12.17 is given by the intersections of the oxygen isobars with the carbon deposition line.

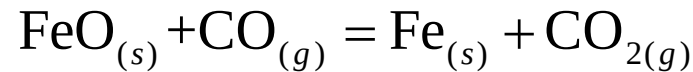
12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

✓ Equilibria such as



can readily be presented on plots such as that in Figure 12.17.

✓ For example, for



$$\Delta G^\circ = -22,800 + 24.26T \text{ J}$$

and thus, the variation of the equilibrium ratio CO_2/CO with temperature is given by

$$\log \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right)_{\text{eq. FeO/Fe}} = \frac{22,800}{2.303 \times 8.3144T} - \frac{24.26}{2.303 \times 8.3144}$$

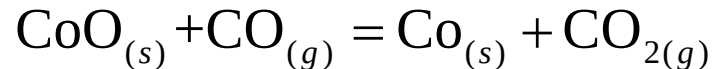
12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ This variation is drawn as the line AB in Figure 12.17, and thus, any gas, the state of which lies above the line AB , is oxidizing with respect to Fe, and states below AB are reducing with respect to FeO.
- ✓ The variation of $\log p_{\text{O}_2(\text{eq.}T, \text{Fe/FeO})}$ with temperature is given by the intersections of the oxygen isobars with the line AB .
- ✓ The temperature at which AB intersects the carbon deposition line is the minimum temperature at which solid FeO can be reduced to solid Fe by graphite, and this is the temperature at which $\text{Fe}_{(s)}$, $\text{FeO}_{(s)}$, $\text{C}_{(s)}$, and the CO-CO_2 atmosphere at 1 atm pressure coexist in equilibrium; that is, it is the temperature at which

$$p_{\text{CO}_2(\text{eq.}C/\text{CO}/\text{CO}_2)} = p_{\text{O}_2(\text{eq.}Fe/\text{FeO})}$$

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ The line CD in Figure 12.17 represents the variation of $\log(p_{\text{CO}_2}/p_{\text{CO}})$ with T for the equilibrium



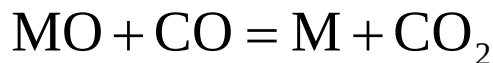
for which

$$\Delta G^\circ = -48,500 + 14.9T \text{ J}$$

and

$$\log\left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}}\right)_{\text{eq. Co/CoO}} = \frac{48,500}{2.303 \times 8.3144T} - \frac{14.9}{2.303 \times 8.3144}$$

- ✓ Since the equilibrium constant K for the reaction



is given by $p_{\text{CO}_2}/p_{\text{CO}(\text{eq. } T, \text{M/MO})}$, a plot of $p_{\text{CO}_2}/p_{\text{CO}(\text{eq. } T, \text{M/MO})}$ versus $1/T$ is a plot of $\log K$ versus $1/T$.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ Figure 12.18 shows the information given in Figure 12.17 as such a plot.
- ✓ With respect to the amount of information which can be obtained from a graphical representation of equilibria in a system, Figure 12.18 is a better representation than is Figure 12.17.
- ✓ Since

$$\left[\frac{d \ln K}{d \left(\frac{1}{T} \right)} \right] = - \frac{\Delta H^\circ}{R}$$

the slope of a tangent to an equilibrium line at the temperature T gives the value of $-\Delta H^\circ/R$.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

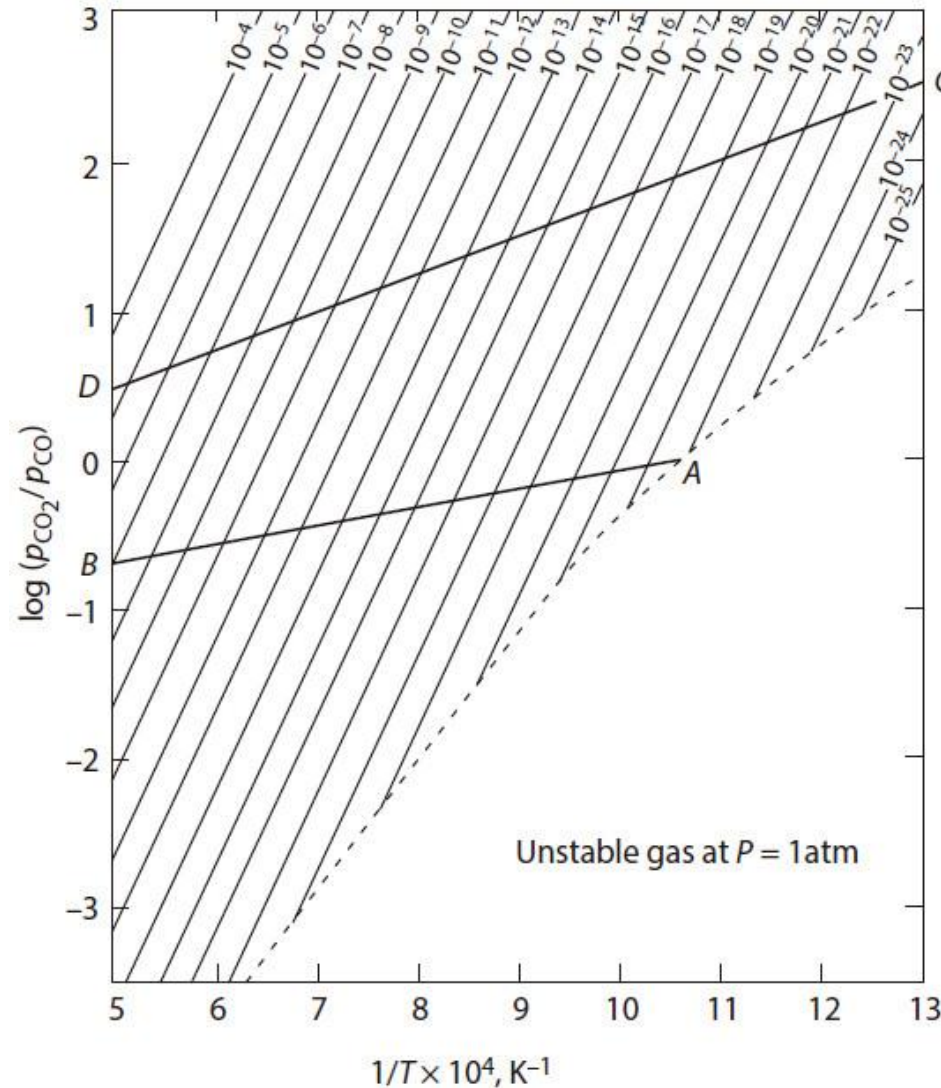


Figure 12.18 Figure 12.17 reproduced as $\log(p_{\text{CO}_2}/p_{\text{CO}})$ vs. $1/T$.

12.7 Graphical Representation of Equilibria in the System Metal-Carbon-Oxygen

- ✓ If $\Delta c_p = 0$, $\log K$ is a linear function of $1/T$. Thus, the slope of the line AB in Figure 12.18 equals $-\Delta H^\circ/R$ for the reaction $\text{FeO} + \text{CO} = \text{Fe} + \text{CO}_2$, and the slope of the line CD equals $-\Delta H^\circ/R$ for the reaction $\text{CoO} + \text{CO} = \text{Co} + \text{CO}_2$.
- ✓ Also, if they are linear, the intercepts of the lines themselves with the $1/T = 0$ axis give the corresponding values of $\Delta S^\circ/R$ for the reactions.
- ✓ From Equation (xi), the slope of any oxygen isobar equals $-\Delta H^\circ/2R$ for the reaction $2\text{CO} + \text{O}_2 = 2\text{CO}_2$, and thus, the oxygen isobars in Figure 12.18 are parallel lines.

Outlines

12.5 The Effect of Phase Transformations

- 12.5.1 Example of the Oxidation of Copper
- 12.5.2 Example of the Chlorination of Iron

12.6 The Oxides of Carbon

- 12.6.1 The Equilibrium $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

12.7 Graphical Representation of Equilibria in the System

Metal-Carbon-Oxygen

12.8 Summary

12.8 Summary

1. The following two facts facilitate a convenient definition of the standard state of species occurring as condensed phases.
 - A pure species occurring as a condensed phase exerts a unique saturated vapor pressure at the temperature T .
 - The dependence of the Gibbs free energy of a condensed phase on pressure (at low pressure) is negligibly small.

This standard state is simply the pure species in its stable condensed state at the temperature T .

12.8 Summary

2. Using this standard state, the equilibrium constant for a reaction involving pure condensed phases and a gas phase can be written in terms of the partial pressures of those species which occur only in the gas phase. For example, for the oxidation of a pure metal to its pure stoichiometric oxide, the equilibrium constant is given by $1/p_{\text{O}_2(\text{eq.}T)}$, where $p_{\text{O}_2(\text{eq.}T)}$ is the unique partial pressure of oxygen required for equilibrium between the metal, its oxide, and the gas phase at the temperature T . This pressure is such that

$$G_{\text{M}}^{\circ} + G_{\text{O}_2}^{\circ} + RT \ln p_{\text{O}_2(\text{eq.}T)} = G_{\text{MO}_2}^{\circ}$$

or

$$\Delta G_T^{\circ} = -RT \ln K = -RT \ln \left(\frac{1}{p_{\text{O}_2(\text{eq.}T)}} \right)$$

12.8 Summary

3. Determination of the equilibrium state of a chemical reaction system requires knowledge of the variation, with temperature, of the standard Gibbs free energy change for the reaction. This relationship can be obtained from thermochemical data—that is, from knowledge of the standard enthalpy and entropy changes at a single temperature (usually ΔH_{298}° and ΔS_{298}°) and the variations, with temperature, of the constant-pressure molar heat capacities of the reactants and products, or it can be determined from knowledge of the variation, with temperature, of the equilibrium constant for the reaction.

12.8 Summary

4. For the oxidation of a pure metal to its pure oxide, experimental measurement of the variation of $p_{\text{O}_2(\text{eq.}T)}$ with temperature gives the variation of ΔG° with temperature as

$$\ln\left(\frac{1}{p_{\text{O}_2(\text{eq.}T)}}\right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} = -\frac{\Delta G^\circ}{RT}$$

If ΔH° and ΔS° are independent of temperature (i.e., if $\Delta c_p = 0$), $\ln K$ is a linear function of $1/T$.

12.8 Summary

5. The fitting of the variation of ΔG° with T to a line produces the so-called Ellingham line for the reaction, and a plot of ΔG° versus T is known as an Ellingham diagram. Ellingham lines plotted on a single diagram for a series of similar reactions (e.g., for the formation of oxides, sulfides, etc.) provide a convenient representation of the relative stabilities of the compounds.

12.8 Summary

6. The addition of p_{O_2} , $p_{\text{CO}}/p_{\text{CO}_2}$ and $p_{\text{H}_2}/p_{\text{H}_2\text{O}}$ nomographic scales to these diagrams facilitates the geometric determination of $p_{\text{O}_2(\text{eq.}T)}$, $p_{\text{CO}}/p_{\text{CO}_2(\text{eq.}T)}$ and $p_{\text{H}_2}/p_{\text{H}_2\text{O}(\text{eq.}T)}$ for, respectively, the equilibria $2\text{M} + \text{O}_2 = 2\text{MO}$, $\text{M} + \text{CO}_2 = \text{MO} + \text{CO}$, and $\text{M} + \text{H}_2\text{O} = \text{MO} + \text{H}_2$.

Homework

1. A mixture of argon gas and hydrogen gas at 1 atm total pressure is passed through a reaction vessel containing a mixture of liquid Sn and liquid SnCl_2 at 900 K. The composition of the gas leaving the vessel is 50% H_2 , 7% HCl , and 43% Ar . Has equilibrium been attained between the gas phase and the liquid phases in the vessel?
2. Fe and FeO are in equilibrium with a gas mixture of composition 71.8% CO –28.2% CO_2 at 1273 K. Which of the two solid phases disappears if the composition of the gas is held constant and the temperature of the system is decreased?

Homework

3. Calculate the vapor pressure of Mg exerted at 1400° C by the system in which the following reaction equilibrium is established.



4. One gram of CaCO_3 is placed in an evacuated rigid vessel of volume 1 liter at room temperature, and the system is heated. Calculate (a) the highest temperature at which the CaCO_3 phase is present, (b) the pressure in the vessel at 1000 K, and (c) the pressure in the vessel at 1500 K. The molecular weight of CaCO_3 is 100.

Thermodynamics of Materials & Phase transformations

The End

